FIRMWARE ARCHITECTURE DOCUMENT

Technical Documentation:

1. Abstract

This document outlines the firmware architecture for the **VitalSphere** IoT edge node. Unlike traditional non-invasive monitoring systems, VitalSphere leverages an ESP32 microcontroller to perform real-time **Edge Computing**. Key innovations include a Single-Layer Perceptron for adaptive sensor calibration, a quadratic regression model for SpO2 accuracy, and a novel **Neural Fatigue Index** derived from sensor fusion. This architecture prioritizes low latency, data integrity, and autonomous operation.

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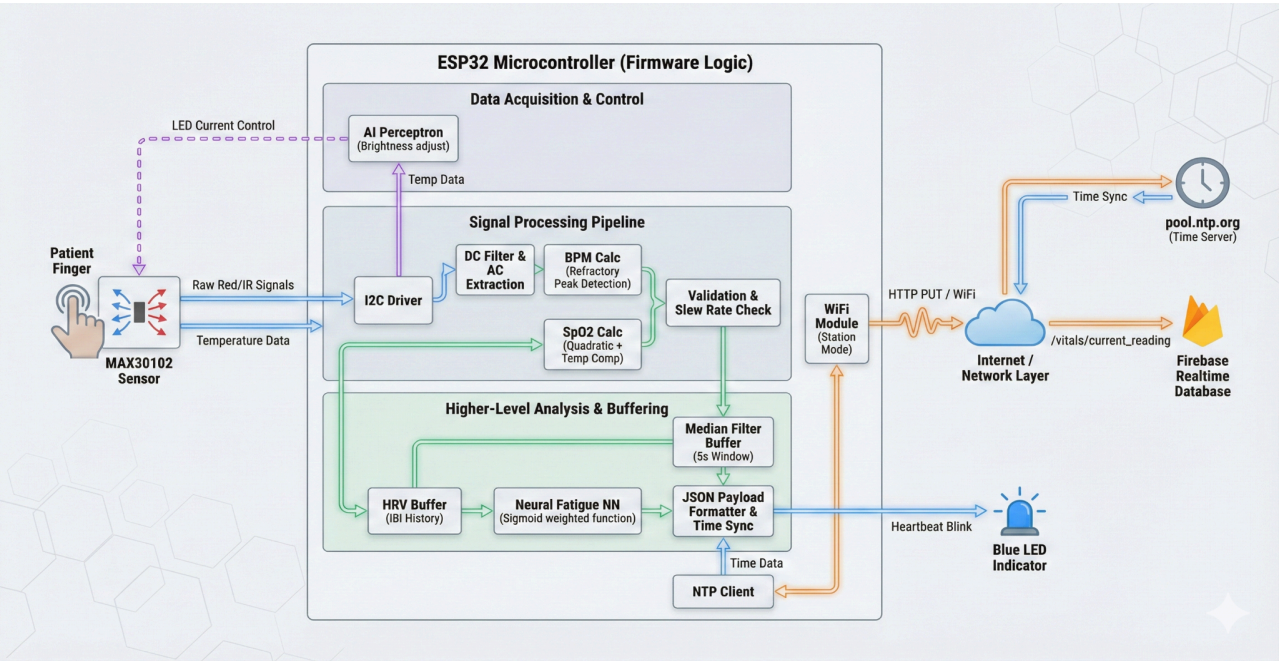
2. System Overview & Architecture



The system is designed as a linear, deterministic pipeline (**"Flow-Wise" Architecture**) that transforms raw photoplethysmogram (PPG) signals into actionable health insights before transmission.

Data Flow Pipeline

The firmware operates in five distinct stages:

`Acquisition` -> `Signal Conditioning` -> `Feature Extraction` -> `Inference` -> `Telemetry`

1. **Acquisition:** The ESP32 polls the MAX30102 sensor (Red/IR/Temp) via I2C at 100Hz.
2. **Conditioning:** Digital filters remove DC offsets (gravity/tissue components) and high-frequency noise.

3. **Extraction:** Peak detection algorithms identify R-peaks for BPM, while AC/DC ratios are calculated for SpO2.
4. **Inference:** On-device logic calculates the Neural Fatigue score and classifies Hypoxia levels.
5. **Telemetry:** Validated data is buffered and pushed to the Firebase Realtime Database via HTTPS.

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3. Core Algorithms & Innovations

A. Adaptive Brightness (Perceptron Loop)

To prevent sensor saturation and thermal injury ("burn-in") on the user's skin, we implement a closed-loop control system using a Single-Layer Perceptron.

- **Objective:** Maintain optimal Signal-to-Noise Ratio (SNR) while minimizing die temperature.
- **Logic:**
  LED Current = (Weight * TempDeviation) + Bias
- **Parameters:**
  - **Weight:** -2.0 (Aggressive negative feedback)
  - **Bias:** 50.0 (Baseline safe current)
- **Outcome:** The system autonomously "dims" the LED if the sensor temperature rises, ensuring safety without user intervention.

B. Precision SpO2 with Thermal Compensation

Standard linear approximation (R = AC/DC) is insufficient for clinical accuracy. We utilize a quadratic model adjusted for vasoconstriction (cold fingers).

- **Base Model:** SpO2 = 110 - 14R - 8(R^2)
- **Thermal Offset:** If the sensor temperature is below 30°C, a compensation factor is applied to account for reduced blood flow:
  Final SpO2 = Raw SpO2 + (0.5 * (30 - Temperature))

C. Neural Fatigue Index (NFI)

A proprietary metric designed to quantify mental and physical exhaustion by fusing three distinct biological vectors:

1. **Vector A (HRV):** Heart Rate Variability (SDNN) - Indicator of Autonomic Nervous System stress.
2. **Vector B (SpO2):** Blood Oxygenation - Indicator of physiological capacity.
3. **Vector C (BPM):** Cardiovascular load.

These vectors are normalized and passed through a weighted **Sigmoid Activation Function**:

NFI = (1 / (1 + e^(-weighted_sum))) * 100%

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4. Signal Integrity & Noise Reduction

To ensure reliability in a hackathon demo environment (where motion is frequent), the firmware employs a multi-stage software filter:

- **DC Removal Filter (Alpha = 0.96):** A high-pass filter that eliminates the static component of the signal (skin/bone absorption), isolating the pulsatile AC component.
- **Slew Rate Limiter:** Any beat-to-beat change exceeding **±30 BPM** is flagged as a motion artifact and discarded.
- **Median Buffering:** A 5-second sliding window collects samples. The system uploads the **Median** value (not the Mean) to effectively filter out transient spikes caused by finger movement.

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5. Hardware Specifications

Component	Specification / Role
Microcontroller	**ESP32-WROOM-32** (Dual Core, 240MHz, WiFi/BLE)
Bio-Sensor	**MAX30102** (High-Sensitivity Pulse Oximeter & Heart Rate)
Visual Interface	Blue LED (GPIO 2) - Syncs with R-Peak detection
Communication	I2C Protocol (400kHz Fast Mode)
Power Supply	3.3V Regulated via USB/LiPo

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6. Cloud Connectivity (JSON Structure)

Data is transmitted asynchronously to maintain sensor polling speed. The payload is formatted as a JSON object:

JSON

```
{
  "timestamp_ms": 1700001234567,     // Unix Epoch for Time-Series Analysis
  "BPM": 72,                         // Median BPM
```

```
  "SpO2": 98.5,                    // Temperature-Compensated SpO2
  "Hypoxia_Status": "Normal",        // Clinical Classification
  "Fatigue_Percent": 14,            // Neural Fatigue Index output
  "IR_Raw": 125000                 // Raw signal for validation
}
```

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7. Future Scope & Commercial Viability

Phase 1: Productization (Hardware)

- **PCB Integration:** Migration from prototyping modules to a custom 4-layer PCB using SMD components to achieve a wearable form factor (Watch/Ring).
- **Casing:** 3D-printed enclosure with light-blocking baffles to eliminate ambient light interference.

Phase 2: Sensor Fusion

- **Galvanic Skin Response (GSR):** Integration of skin conductance sensors to correlate "Neural Fatigue" with sweat gland activity for highly accurate stress detection.
- **IMU (Accelerometer):** Adding an MPU6050 to detect falls in elderly patients and intelligently pause scanning during heavy physical activity.

Phase 3: Advanced AI (Federated Learning)

- **On-Device Training:** Moving from hardcoded weights to on-device learning. The device will learn the user's specific baseline HRV over 7 days, allowing for personalized anomaly detection rather than generic thresholds.